

STRATEGIC FACILITY DOSSIER: RELIANCE JAMNAGAR SUPER REFINERY

CODE: FAC-GUJ-001

Sector: Refinery | Operator: Independent | Sovereign Territory: Gujarat, India

Generated: 2026-09-01 16:48 UTC

WINDOW DETECTIONS (30d) 2 2 Active Operational Days	PEAK RADIATIVE POWER 363.0 MW Mean: 182.2 MW	90-DAY BASELINE ($\mu \pm \sigma$) 165.0 MW ± 35.0 MW (N=48)	ACTIVITY TREND INCREASING Max Streak: 1 Days
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Temporal Radiative Intensity (Peak FRP MW per satellite detection pass)



1. Grounded Tactical Intelligence Assessment

OBSERVED	- Industrial plant: Reliance Jamnagar Super Refinery (Refinery / Independent Operator). - Location coordinates: (22.3512°N, 69.8524°E) in Gujarat, India. - Total recorded thermal events in 30-day window: 2 across 2 distinct active operational days. - Observed peak radiative power: 363.0 MW (Mean: 182.2 MW).
DERIVED	- Empirical 90-day baseline envelope: Mean = 165.0 MW, Std = ± 35.0 MW (Sample size N = 48). - Peak Z-Score anomaly intensity: $Z = +5.66\sigma$ relative to established sovereign baseline. - Operational flaring temporal trend: INCREASING across the 30-day observation window.
MODELLED	- Primary combustion classification: IND_FIRE. - Dominant anomaly classification distribution: {'CRITICAL': 1, 'NORMAL': 1}.
UNKNOWN	- Internal plant process adjustments (maintenance flare vs unannounced blowdown) cannot be distinguished by satellite radiometry alone. - Sub-resolution optical emissions (< 375m VIIRS pixel footprint) may fall below radiative detection threshold.

2. Monitored Combustion Detections (30-Day Registry)

Event ID	Detection Time (UTC)	Peak FRP	Classification	Severity Tier	Z-Score
EVT-IN-GUJ-0001	2026-08-31 07:31	363.0 MW	IND_FIRE	CRITICAL	+5.66 σ
EVT-IN-NON-7426D182	2026-08-26 17:16	1.5 MW	OTHER_UNCERTAIN	NORMAL	-4.67 σ

3. Sovereign Geospatial & Land-Use Verification (5km Buffer)

Centroid Coordinates: 22.35120°N, 69.85240°E	Built-up Industrial: 74.5%	Barren Soil / Open Land: 14.2%
Sovereign State: Gujarat, India (Survey of India Verified)	Vegetation & Tree Cover: 8.5%	Water Bodies: 2.8%

PROVENANCE SHA-256: de1c4bce090f119964a694492551da176bef0493facb8ac57bedf646460a860f

Grounded Analytical Report · ThermoTrace AI Platform (PS 26162 Deadlock) · CPCB-NTRO Compliance

OFFICIAL SOVEREIGN RECORD

Verified Sensor Telemetry